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Education

Texas A&M University

Expected Graduation: May 2028

Bachelor of Science: Computer Science

Location: College Station, TX

- Student at Texas A&M University majoring in Computer Science
- Relevant Coursework: Data Structures and Algorithms, Discrete Structures for Computing, Program Design and Concepts, Computation (Python), Statistics, Linear Algebra

Relevant Skills

Frameworks & Languages: Python, C++, FastAPI, SQL, HTML/CSS, TensorFlow, PyTorch, XGBoost, TypeScript, pytest, Vitest

Tools & Platforms: GitHub, React, Jupyter, Google Colab, VS Code, Remix IDE, Cursor, Claude Code, Vercel, Node.js, Docker, Azure, REST APIs, Git, Github Actions, Model Context Protocol (MCP)

Work Experience

Maroon Fund

January 2026 – Present

Quantitative Developer

College Station, TX

- Backtested combined buy/sell signal pipeline (NVDA, AMZN, META) — day-by-day trade simulation, trade log sanity checks, standardized local CSV inputs feeding 20+ performance metrics (Sharpe/sortino, drawdown, win rate, profit factor).
- Extended backtest stack with benchmark analytics versus S&P 500 (alpha, beta, volatility, benchmark return) on 3-tier data fallback plus CLI improvements for repeatable runs at a student-managed quant fund with \$70K+ AUM.

Arytic

July 2025 – November 2025

Software Development Intern (Remote)

Austin, TX

- Implemented python-based parsing engines to parse and segment PDF resume content into structured JSON outputs, extracting contact fields and matching against a 220+-term skills taxonomy for system interoperability.
- Engineered a semantic-search pipeline in Pinecone, upserting 384-dim BERT-based embeddings (all-MiniLM-L6-v2, Hugging Face Transformers) for cosine-similarity resume retrieval against job-description queries.
- Prototyped keyword- vs. embedding-based retrieval variants to compare candidate-to-role matching quality.

University of North Texas: Smart Electronics Systems Laboratory (Remote)

July 2024 – October 2024

High-School Research Assistant

Denton, TX

- Researched integrating Single Sign-On (SSO) with Decentralized Identifiers (DIDs) to enhance authentication in Software Patch Delivery Systems, analyzing protocols like OAuth 2.0 and DID methods such as DID:ETHR.
- Utilized Ethereum IDE for contract-contract interactions with Solidity, developing a modular framework proposal.
- Developed and deployed a message board smart contract as a decentralized system in Remix's Ethereum IDE, implementing storage and retrieval functions with Solidity fundamentals.

Projects

Amber: AI Infrastructure Simulator | [Source Code](#)

September 2026 – Present

- Built a full-stack simulator (Docker, Azure Container Apps) where users sketch LLM system architectures on a canvas and run simulated traffic through them, projecting p50/p99 latency, utilization, bottlenecks and monthly cost before deployment.
- Modeled self-hosted LLM inference (continuous batching, KV-cache memory limits, speculative decoding) and agentic workloads with rate-limit retries; shipped an MCP server so AI agents (Claude Code, Cursor) can model and simulate a user codebase.
- Validated the discrete-event engine against M/M/1 and M/M/c queueing theory, matching within 0.8% over 100 runs each; 400+ pytest and Vitest tests run in GitHub Actions CI, which flags any API mismatch between frontend and backend.

Open Source Contributions | [GitHub Link](#)

July 2026 – Present

- Landed 3 pull requests (4 commits) across 2 TypeScript and Python developer tooling repos, each shipped with regression tests and reproducible demos.
- Fixed character-rendering bug in token-cost cutting tool, repainting 5x8 bitmap glyph atlas at zero token cost, and calibrated cost-profitability gate with a least-squares fit over logged telemetry, replacing the cost gate's chars-per-token constant.
- Capped an unbounded file-read tool in chaosblade's LLM agent to a 256 KB byte-exact prefix with a truncation notice to prevent context overflow and runaway token cost.

Brittle: Interactive Stock Portfolio Fragility Playground | [Source Code](#)

May 2026 – July 2026

- Deployed a layered full-stack tool collapsing any ticker/weight set into a 0–100 Fragility Score, engineered from correlation, HHI concentration, PCA hidden-factor share (NumPy SVD/PC1), drawdown, volatility, and age-weighted VaR/CVaR across 3 weighted risk branches.
- Modeled single-point-of-failure risk via a 30% PC1-exposure shock with correlation-aware contagion, quantifying loss amplification vs. weight alone; hardened with 40 network-free pytest cases and per-ticker price caching.

Chest X-Ray Pneumonia Localization Architectures | [Source Code](#)

December 2023 – July 2024

- Built and compared CNN architectures (Keras ResNet50V2, InceptionV3) in Google Colab for pneumonia localization/detection on RSNA chest X-ray data, achieving ~97% validation/test accuracy and IoU over 0.700.
- Preprocessed ~30,000 DICOM images with a custom data generator (flipping, interpolation, RGB channel stacking, resizing) to improve model robustness across architectures.